PROJECT REPORT

Employee Resource Management System

Built with Microsoft Power Apps, Dataverse and Power Automate

Technologies: Dataverse | Canvas App | Model-Driven App | Business Rules | Microsoft Forms | Power Automate

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1. Introduction

The Employee Resource Management System is a centralized workforce management solution built on the Microsoft Power Platform. It tracks employees across different resource states, stores employee information in one controlled location, automates update requests through an approval workflow, and gives managers clear visibility into how their teams are allocated.

Organizations often manage employee allocation through spreadsheets, e-mail threads and manual updates. This approach produces outdated data, duplicate records and no reliable audit trail. This project replaces that approach with a low-code solution in which Microsoft Dataverse acts as the single source of truth, and several Power Platform components expose and maintain that data.

1.1 Questions the System Answers

- Which employees are currently assigned to projects?
- Which employees are on the bench?
- Which employees are undergoing training?
- Which resources are available for new assignments?
- How many employees report to each manager?
- How can employee information be updated in a controlled manner?

1.2 Solution Summary

The solution consists of six cooperating components: a Dataverse table that stores the Employee Record data, a Canvas App for employees and managers, a Model-Driven App for administration, Business Rules that enforce data quality, a Microsoft Form for submitting update requests, and a Power Automate flow that routes each request through manager approval before updating Dataverse.

2. Business Problem

Before this system, employee allocation information was maintained informally. The consequences fall into four areas, summarised below.

Problem Area	Observed Challenges
Data Management	Duplicate employee records, outdated information and inconsistent updates across files.
Resource Tracking	Difficulty identifying available employees, limited visibility into bench resources and no tracking of training status.
Approval	No formal update process, unauthorized changes and difficulty auditing who changed what.
Reporting	Time-consuming report generation, limited manager visibility and no centralized dashboard.

2.1 Impact on the Organization

When allocation data is unreliable, managers cannot confidently answer a basic question such as who is free to join a new project. Staffing decisions are delayed, bench time goes unnoticed, and training plans are not aligned with actual demand. A formal, auditable process for updating resource information is therefore as important as the storage of the data itself.

2.2 Proposed Approach

The proposed system tackles each problem area directly. Central storage in Dataverse removes duplicates and inconsistent copies. Four well-defined statuses make resource states unambiguous. An approval-based flow ensures that every change is reviewed by a manager. Finally, the Canvas App provides an instant dashboard so that reports no longer need to be compiled by hand.

3. Project Objectives

3.1 Employee Management

Maintain employee information in a centralized location so that every user works from the same, current data.

3.2 Resource Allocation Tracking

Track every employee across four statuses: Project Assigned, Training, Bench and Available.

3.3 Manager Visibility

Provide manager-wise summaries that include the following figures:

- Total Employees
- Assigned Employees
- Benched Employees

- Training Employees
- Available Employees

3.4 Automated Updates

Enable employees to submit update requests through a form and process them through an approval workflow, rather than editing data directly.

3.5 Data Validation

Implement business rules that automatically enforce data quality standards, such as showing the Project Name field only when relevant and setting values automatically.

3.6 Scope

The scope covers employee records, manager information, skills, status and project details. Advanced analytics, Teams notifications and role-based security are identified as future enhancements and are not part of the current release.

4. Use Cases

The system serves three types of users. Each use case below describes who performs it and what the system provides.

4.1 Actors

Actor	Role
Employee	Searches employee information in the Canvas App and submits update requests through the Microsoft Form.
Manager	Views team summaries and approves or rejects update requests.
Administrator	Creates, edits and deletes employee records in the Model-Driven App.

4.2 Use Case Summary

ID	Use Case	Actor	Outcome
UC-01	Search employees by name	Employee, Manager	Matching employee cards are displayed
UC-02	Filter employees by status	Employee, Manager	List shows only Project Assigned, Training, Bench or Available staff
UC-03	View manager-wise summary	Manager	Totals for assigned, benched, training and available staff
UC-04	Submit update request	Employee	Request is sent for approval
UC-05	Approve or reject request	Manager	Dataverse is updated or a rejection e-mail is sent
UC-06	Manage employee records	Administrator	Record is created, updated or deleted in Dataverse

ID	Use Case	Actor	Outcome
UC-07	Auto-show Project Name	Administrator	Project Name field appears when the project condition is met
UC-08	Auto-set joining value	Administrator	Field value is set automatically by the Joining Date rule

4.3 Detailed Use Case: Submit and Approve an Update Request

Item	Description
Use Case ID	UC-04 and UC-05
Actors	Employee, Manager, System (Power Automate)
Precondition	The employee record exists in Dataverse and the employee has access to the Microsoft Form.
Main Flow	1. Employee fills in the form and submits it. 2. The flow retrieves the response. 3. An approval request is sent to the manager. 4. The manager approves. 5. The flow finds the employee record and updates it. 6. A confirmation e-mail is sent.
Alternate Flow	The manager rejects the request. A rejection e-mail is sent and no data is changed.
Postcondition	Approved changes are visible in both the Canvas App and the Model-Driven App.

4.4 Detailed Use Case: View Team Utilization

Item	Description
Use Case ID	UC-03
Actor	Manager
Precondition	Employee records with manager and status values exist in Dataverse.
Main Flow	1. Manager opens the Canvas App. 2. Manager selects Managers on the Home screen. 3. The Manager Information screen lists each manager with total, assigned, benched, training and available counts.
Postcondition	The manager can identify team utilization, availability and training needs.

5. Technology Stack

The project uses components of the Microsoft Power Platform. Each component has a distinct responsibility, and together they provide an end-to-end low-code solution.

Component	Role in the Solution
Microsoft Dataverse	Central database storing employee details, manager information, skills, status information and project details.
Power Apps Canvas App	Customized, mobile-friendly interface for employees and managers: search, status filtering, resource dashboards and manager summaries.
Power Apps Model-Driven App	Administrative interface for creating, updating and managing Dataverse records.
Business Rules	No-code automation for field visibility, field updates and data validation.
Microsoft Forms	Front end for employees to submit update requests without direct database access.
Power Automate	Form processing, approval workflow, Dataverse updates and e-mail notifications.

5.1 Why Low-Code?

A low-code platform shortens development time, keeps the solution maintainable by non-developers, and integrates naturally with Microsoft 365 services such as Outlook, Approvals and Forms. Because all components share the same Dataverse back end, a change made through any channel is immediately reflected everywhere else.

6. Data Architecture

The Employee Record table in Dataverse is the primary data source for the entire system. Every screen, form and flow reads from or writes to this table.

6.1 Employee Record Table

Field	Description
Employee Name	Employee full name
Employee ID	Unique identifier for each employee (for example EMP001)
Email	Employee e-mail address
Manager	Reporting manager
Current Status	Resource status: Project Assigned, Training, Bench or Available
Skills	Technical skills (for example Java, Spring Boot, SQL)
Joining Date	Date on which the employee joined
Project Name	Assigned project

6.2 Employee Status Categories

Status	Meaning	Purpose
Project Assigned	Employees actively working on projects	Revenue-generating, fully allocated workforce
Training	Employees undergoing training programs	Skill enhancement and upskilling
Bench	Employees not assigned to active projects	Awaiting allocation; available for future opportunities
Available	Resources ready for project assignment	Immediate staffing and short-term allocation

The Employee ID is intended to be unique, which allows the update flow to locate the correct record reliably. The Model-Driven App list shown later in this report contains 200 employee records.

7. Canvas App

The Canvas App was developed to provide a simple, user-friendly interface for managers and employees. It is designed for a phone-sized layout, which gives a minimal learning curve and a mobile-friendly experience.

7.1 Home Screen

The Home screen, titled Resource Management, is the landing page. It displays the prompt "Select the information you want to view" and offers two navigation buttons: Managers and Employees.

7.2 Employee Screen

The Employee screen lets users search employees by name and filter them by status using a drop-down (Project Assigned, Training, Bench or Available). Each employee card shows the name, Employee ID, status and skills, and a back button returns to the Home screen. In the screenshot the filter is set to Project Assigned, and the list shows employees such as Rahul Singh (EMP016) with Java, Spring Boot and SQL skills, and Nisha Singh (EMP015) with Power BI, Excel and DAX skills. The benefits are fast resource identification, an improved user experience and better resource visibility.

7.3 Manager Dashboard

The Manager Information screen provides manager-wise resource statistics. For each manager it displays Total Employees, Assigned Employees, Benched Employees, Training Employees and Available Employees.

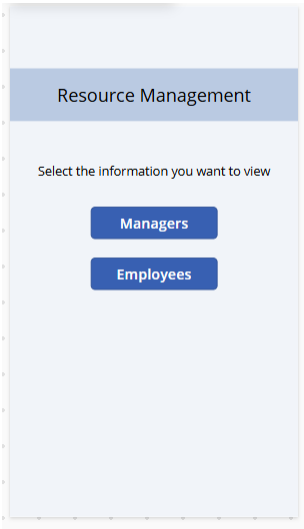


Figure 1: Home screen

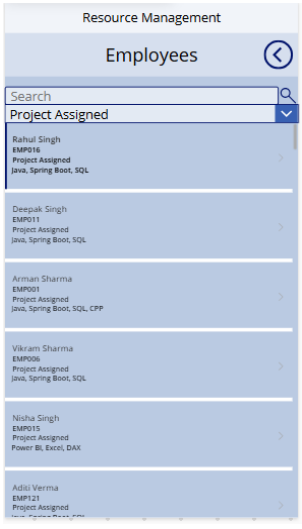


Figure 2: Employee screen



Figure 3: Manager Information

The data in the Manager Information screenshot shows how quickly a manager's situation can be read. For example, Rajesh Sharma has 39 employees, all assigned; Amit Sethi has 40 employees, all available; and Pooja Arora has 41 employees, of whom 40 are in training and 1 is assigned. Managers can therefore immediately identify team utilization, team availability and training requirements.

8. Model-Driven Application

A Model-Driven App was developed to provide data management capabilities directly on top of Dataverse. Where the Canvas App is optimized for viewing, the Model-Driven App is optimized for administration.

8.1 Employee Records View

The Active Employee Records view lists the records in the Employee Record table. The command bar provides New, Delete and Refresh actions along with a Share menu, and users can filter by keyword, sort by column and select records, which gives centralized administration and rapid data access. The footer confirms that the view contains 200 rows.

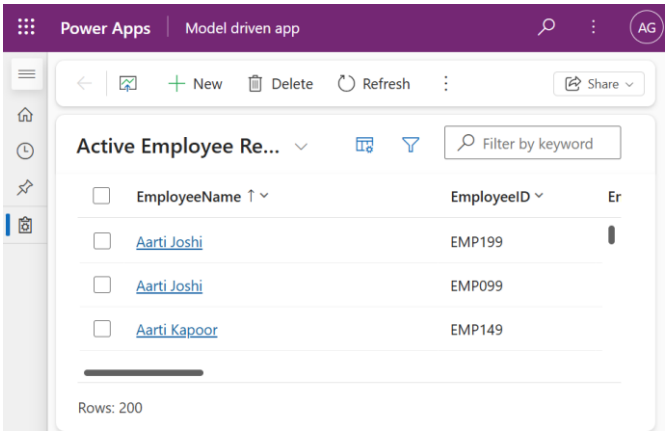


Figure 4: Model-Driven App: Active Employee Records view

8.2 Employee Form

The Employee form is used to create and edit employee records. The screenshot shows the New Employee Record form with the General tab, the owner, and the first fields (EmployeeName, EmployeeID and Email). The command bar offers Save, Save & Close, New and Flow options.

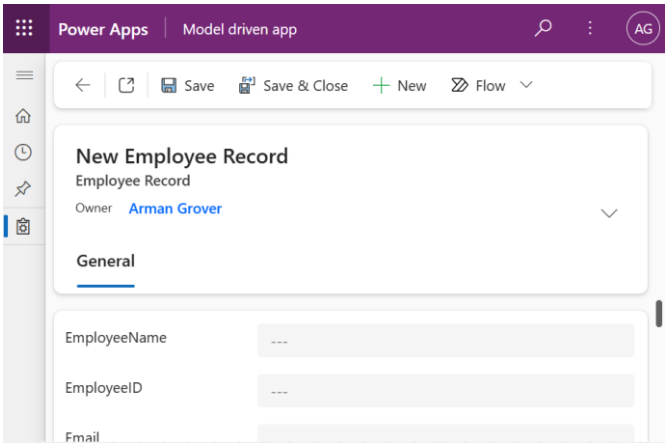


Figure 5: Model-Driven App: New Employee Record form

The form captures Employee Name, Employee ID, Email, Skills, Manager, Current Status and Project Details, giving structured data entry and consistency.

9. Business Rules Implementation

Business Rules automate the behavior of the form without writing code. Each rule consists of a condition and one or more actions that run when the condition is true. Two rules were implemented in this project.

9.1 Business Rule 1: Project Visibility

This rule contains a Condition named "Project check" followed by a Set Visibility action named "Visible Project name". When the condition is satisfied, the Project Name field is shown on the form.

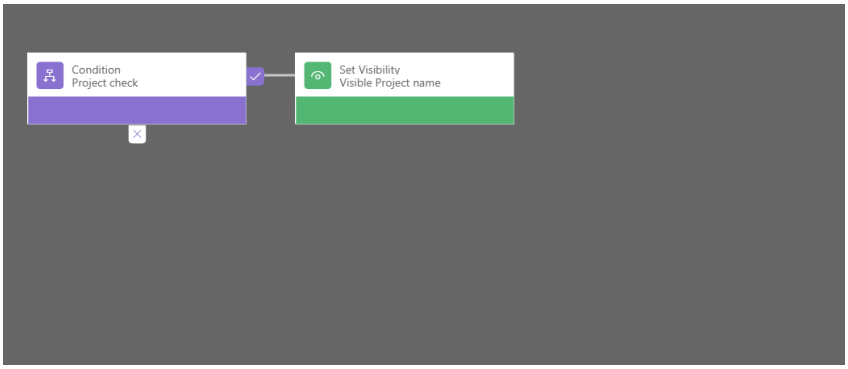


Figure 6: Business rule: Project check and Set Visibility

- Cleaner forms, because the field appears only when it is needed
- Better user experience
- Dynamic field management

9.2 Business Rule 2: Joining Date Rule

This rule contains a Condition named "Joining date condition" followed by a Set Field Value action named "Set joining". When the condition is met, a field value is set automatically from the employee joining date information.

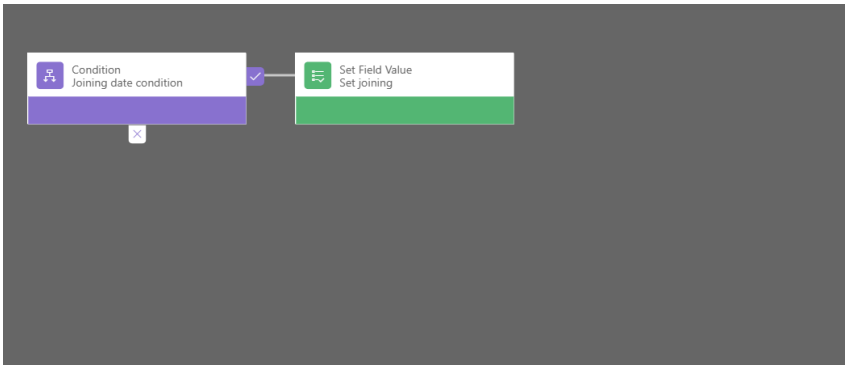


Figure 7: Business rule: Joining date condition and Set Field Value

- Reduced manual entries
- Improved consistency
- Better data quality

Both rules follow the same structure. The green check path on the condition runs the action, while the cross path represents the case in which the condition is not met.

10. Microsoft Forms Integration

A Microsoft Form titled "Update employee resource information" lets employees submit resource update requests. It gives employees a standard way to request a change without any direct access to the database.

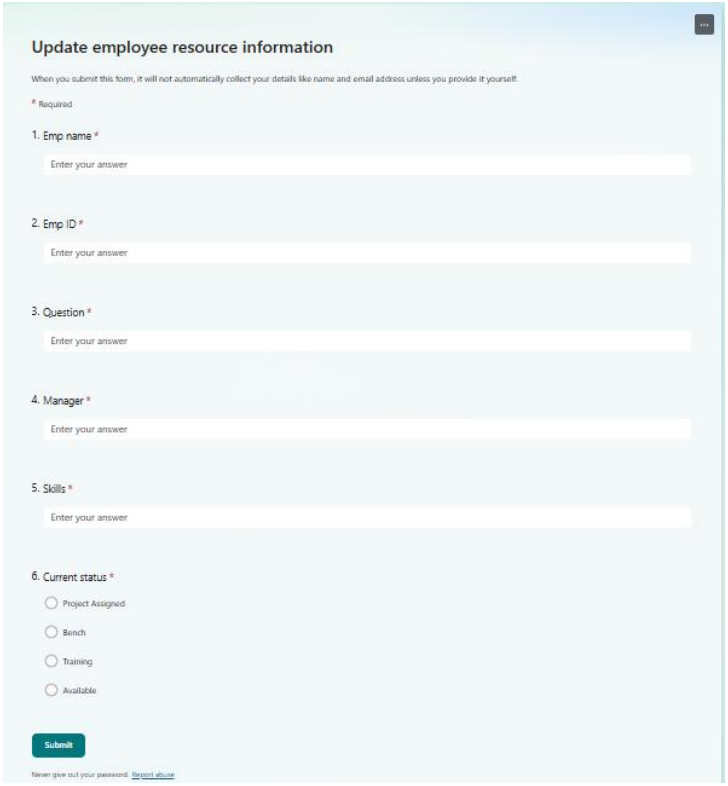


Figure 8: Microsoft Form for update requests

10.1 Information Collected

No.	Question	Type / Notes
1	Emp name	Required, text answer
2	Emp ID	Required, text answer
3	Question	Required, text answer describing the update request
4	Manager	Required, text answer (the manager who will approve)
5	Skills	Required, text answer
6	Current status	Required, single choice: Project Assigned, Bench, Training, Available

Because the Current status question is a fixed choice, the submitted value matches the Dataverse status categories. Every question is marked as required, which prevents incomplete submissions. The benefits are easy information submission, no direct database access and a standardized update process.

11. Power Automate Workflow

An approval-based flow was implemented in Power Automate. It starts when a form response is submitted, asks the manager for approval, and then either updates Dataverse or notifies the employee of the rejection.

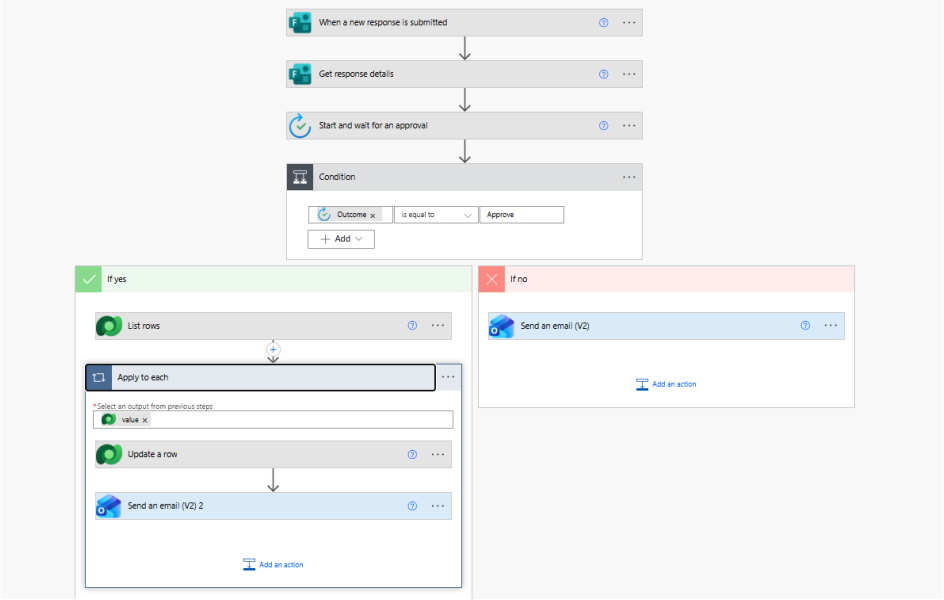


Figure 9: Power Automate approval flow

11.1 Flow Steps

1. Trigger: "When a new response is submitted" starts the flow when an employee submits the Microsoft Form.
2. "Get response details" retrieves the answers of the submitted response.
3. "Start and wait for an approval" sends an approval request to the manager and pauses the flow until a decision is made.
4. Condition: checks whether the approval Outcome is equal to Approve.
5. If yes: "List rows" locates the employee record in Dataverse, "Apply to each" loops over the matching rows, "Update a row" writes the new values and "Send an email (V2) 2" sends a confirmation.
6. If no: "Send an email (V2)" sends a rejection e-mail to the employee.

11.2 Design Notes

The "Apply to each" loop appears because the List rows action returns a collection, even when only one employee matches. Filtering the list by Employee ID ensures that only the intended record is changed. Because Dataverse is updated only on the approval branch, no change reaches the data without manager consent, which provides the governance and auditability that the manual process lacked.

12. End-to-End User Journey

Actor	Actions
Employee	Opens the Microsoft Form, submits the update request and waits for the decision.
Manager	Receives the approval notification, reviews the request and approves or rejects it.
System	On approval: finds the employee record, updates Dataverse and sends a confirmation e-mail. On rejection: sends a rejection e-mail.
Any user	Sees the updated data in the Canvas App and in the Model-Driven App.

12.1 Data Flow

Data moves in one direction for updates: Microsoft Form, then Power Automate, then manager approval, then Dataverse. Both apps read from Dataverse, so a single approved update is visible in the Canvas App dashboards, the Employee list and the Model-Driven views without any additional synchronization.

12.2 Sample Test Scenarios

#	Scenario	Expected Result
1	Employee changes status from Bench to Training and the manager approves	Dataverse record shows Training; confirmation e-mail is sent
2	Manager rejects the request	Record is unchanged; rejection e-mail is sent
3	Employee filters by Bench in the Canvas App	Only benched employees are listed
4	Manager Information screen is opened after an approved update	Manager counts reflect the new status
5	New record is created in the Model-Driven App	Record saved and visible in the Active Employee Records view

13. Key Features and Benefits

13.1 Key Features

- Centralized employee repository
- Employee search and status filtering
- Manager dashboard
- Resource allocation tracking
- Business rule automation
- Approval workflow with e-mail notifications
- Dataverse integration
- Model-driven administration
- Microsoft Forms integration

13.2 Benefits Achieved

Category	Benefits
Operational	Reduced manual work, faster updates and improved data accuracy.
Manager	Resource visibility, utilization tracking and team summaries.
Organizational	Better planning, improved governance and standardized processes.

14. Future Enhancements

Enhancement	Description
Power BI Integration	Advanced workforce analytics and dashboards.
Microsoft Teams Integration	Automated Teams notifications for approvals and decisions.
Security Roles	Department-specific data access.
AI Integration	Resource recommendation and workforce forecasting.
Self-Service Portal	Employee-managed profile updates.

15. Conclusion

The Employee Resource Management System demonstrates an end-to-end Microsoft Power Platform solution that combines Dataverse, a Canvas App, a Model-Driven App, Business Rules, Microsoft Forms and Power Automate.

The application centralizes employee information, automates approval processes, improves resource visibility and provides a scalable foundation for workforce management. It reduces manual effort, improves governance and enables managers to make informed resource allocation decisions through a modern low-code platform.